

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed wafer acceptance testing (WAT) capabilities to support new and existing FAB process developments, improving test structure library management for novel technologies.
- Resolved non-genuine yield loss at WAT by performing root cause analysis on device and process metrics, implementing corrective actions and robust solutions.
- Drove continuous improvement activities for existing volume manufacturing processes, focusing on software enhancements and test time reduction at WAT to meet FAB output targets.
- Managed test structure libraries for each process and designed new test structures for emerging technologies, ensuring alignment with the development technology roadmap.
- Owned the performance of a key Fab Process as a Unit Device and process expert, addressing process and yield issues affecting key metrics.
- Developed a coherent view of manufacturing processes from start to finish, identifying critical nodes and ensuring sufficient controls across the production flow.
- Containerized Python-based data processing scripts for WAT results analysis using **Docker** and deployed them to an **AWS** environment, enabling scalable data ingestion and reporting.

SKILLS

Programming Languages – Java, Python, C, SQL, R

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Data Analysis & BI – MS Excel, PowerPoint, Power BI, Tableau, DAX, Power Query

Engineering & Testing – Wafer Acceptance Testing (WAT), Device Engineering, Process Controls, Yield Debug, Test Methodologies, Root Cause Analysis, Continuous Improvement

Software Development – Spring Boot, REST APIs, Git, Jenkins, Jira, Confluence

PROJECTS

Automated WAT Data Analysis Pipeline (HCL Technologies)

- Engineered a **Python** pipeline to process and analyze **Wafer Acceptance Testing (WAT)** data, automating the extraction of key performance indicators from raw test logs.
- Integrated **SQL** queries to validate test data against established process parameters and identify anomalies, reducing manual data validation effort by approximately 40%.
- Developed interactive **Power BI** dashboards to visualize WAT results, process trends, and yield deviations for manufacturing engineers, facilitating faster decision-making.

FAB Process Control & Yield Improvement Initiative (HCL Technologies)

- Collaborated with operations teams to define and implement enhanced **process controls** for a critical semiconductor fabrication process, aiming to reduce variability.
- Analyzed historical yield data using **Python** and statistical methods to pinpoint root causes of recurring excursions, contributing to a targeted 15% reduction in specific defect types.
- Designed and documented updated troubleshooting runbooks for yield-related issues, improving incident response time for the production support team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)